

The module measures 16mm x 16mm and supports all long-term evolution for machines (LTE-M) and narrow-band Internet of Things (NB-IoT) bands and offers a radio frequency



(RF) output power of 23dBm. It can operate on a 2G network for added flexibility.

The module is suited for various use cases, including small asset trackers like pet and personal trackers, micro-mobility devices, and luggage tags. However, its versatility extends to other applications such as alarm systems, vending machines, and stolen vehicle recovery. The compact module achieves a 40% reduction in footprint compared to similar modules, allowing the potential integration of larger antennas for improved RF performance or accommodating larger batteries. Additionally, the module supports Global Navigation Satellite System (GNSS) AT commands, is compatible with GNSS bundling, and integrates effortlessly with other GNSS modules for improved performance and energy efficiency.

u-blox

<https://www.u-blox.com/en>

Screwless terminal blocks

Omni Pro Electronics has introduced the EB4862 series of easy-wiring 'screwless' terminal blocks. These blocks offer a user-friendly push-button solution. With a finger-friendly lever



that requires minimal force for operation, these terminal blocks come in a modular

design with 30 positions and can be colour-coded to prevent wiring errors. The innovative push-to-connect system allows for easy insertion of both solid and stranded wires without the need to depress the lever. The terminal blocks feature a 37° angled wire entry, providing a clear view of the wire entry and lever on the same 45° plane, which aids in rapid field wiring. Constructed

for durability, the terminal blocks utilise UL 94V-0 polyamide PA66 insulator body material and brass contact block material with tin plating. They have an operating temperature range of -40°C to 119°C and are suitable for various applications, including irrigation controls, Wi-Fi controllers, thermostats, pool/spa controls, keypads, sensors, detectors, and alarms. With a voltage rating of 300V AC and a current rating of 8A, the terminal blocks can accept wire sizes from 14 to 22AWG, providing a versatile and reliable wiring solution for a wide range of electrical systems.

Omni Pro Electronics

www.omnipro.net

BOARDS

Arduino Nano ESP32

Arduino has launched the Arduino Nano ESP32, the first Arduino board built around an Espressif ESP32 microcontroller. The board features a NORA-W106 module from u-blox with a USB-C connector, 16MB flash, and compatibility with MicroPython



and the Arduino IoT Cloud. This innovative board couples the user-friendly and adaptable environ-

ment of the Arduino with the energy-efficient potential of the ESP32-S3 system-on-a-chip microcontroller. The applications are limitless in home automation, gaming, and education. Designed with a nano form factor, this compact board is ideal for standalone projects due to its small footprint. Powered by the ESP32-S3 microcontroller, it provides Arduino-supported Wi-Fi and Bluetooth. The board also supports HID, allowing emulation of devices like keyboards, enhancing computer interaction possibilities. The Nano ESP32 provides immediate Arduino Cloud support. Sharing information across devices, integration with Alexa and much more is possible

with this Arduino board.

Arduino

<https://www.arduino.cc>

An ARM-based single-board computer

ASUS IoT has announced the Tinker Board 3N series, an ARM-based single-board computer (SBC) for the Industrial Internet of Things (IIoT). The next unit of computing (NUC) sized



SBC series supports Linux Debian, Yocto, and Android, offering an option for various IIoT projects.

The thermal design ensures efficient operation, while its durable build and enhanced computing performance make it suitable for smart manufacturing applications in the IoT revolution. The board offers enhanced GPU performance for graphics processing with low energy usage. It is suitable for IoT gateway, human-machine interfacing (HMI), and factory automation. The board includes mechanical design improvements for suitable embedded application use. Its compact NUC-scale dimensions allow for deployment in space-constrained areas and flexible system integration. The board operates smoothly within a temperature range of -40 to 85°C, meeting the demands of industrial automation.

ASUS IoT

<https://iot.asus.com>

Development board for automotive gateway systems

Renesas Electronics Corporation has launched a development board, R-Car S4 starter kit. The board is designed for automotive gateway systems and is highly affordable. The development board is built around the company's R-Car S4 system on chip (SoC), which boasts computing capabilities and a



wide range of communication features. These features